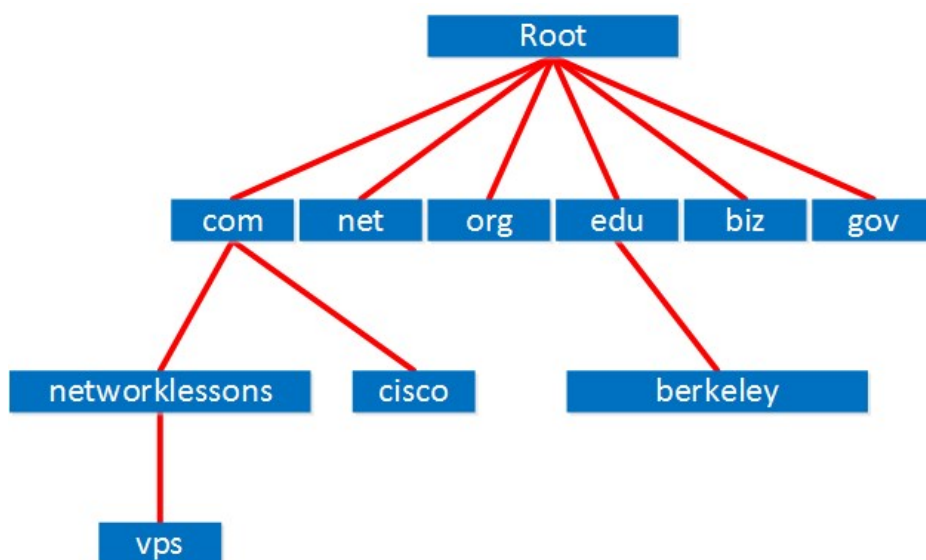


Introduction to DNS

DNS (Domain Name System) is a network protocol that we use to find the IP addresses of hostnames. Computers use IP addresses but for us humans, it's more convenient to use domain names and hostnames instead of IP addresses. If you want, you could visit networklessons.com by going directly to IP address 95.85.36.216, but typing in the domain name networklessons.com is probably easier.

DNS is **distributed and hierarchical**, there are thousands of DNS servers, but none of them has a complete database with all hostnames / domain names and IP addresses. A DNS server might have information for certain domains but might have to query other DNS servers if it doesn't have an answer.

There are 13 root name servers that have information for the generic top level domains like com, net, org, biz, edu or country specific domains like uk, nl, de, be, au, ca, and such. Take a look at the image below:



At the top of the DNS hierarchy are 13 root name servers that contain name server information for the top level domain extensions. For example, a name server for .com will have information on networklessons.com, but it won't know anything about networklessons.org. It will have to query a name server that is responsible for the org domain extension to get an answer.

Below the top level domain extensions you will find the **second level domains**. Here's where you find the domain names like networklessons, Cisco, Microsoft, etc.

Further down the tree, you can find hostnames or subdomains. For example, vps.networklessons.com is the hostname of the VPS (virtual private server) that runs this website. An example of a subdomain is tools.cisco.com where vps.tools.cisco.com could be the hostname of a server in that subdomain.

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Between each DNS “record” we use a period character (.) and officially we also have to use a period character for the root, but almost nobody writes or prints it. Take a look at the two examples below:

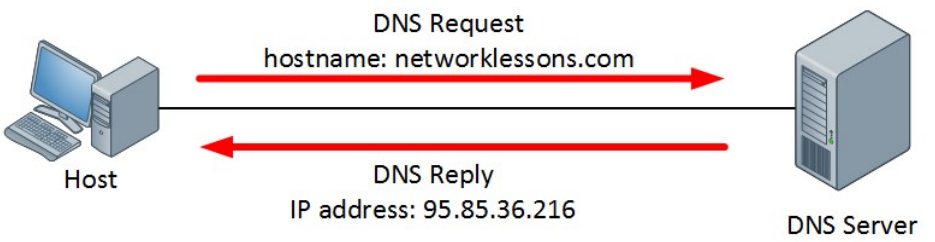
- vps.networklessons.com.
- vps.networklessons.com

Take a close look at those examples above; the first one has a trailing period character that indicates the root of the DNS hierarchy. Writing down a hostname with its complete domain name like we did above is called an **FQDN (Fully Qualified Domain Name)**.

Here’s a summary of what I just explained:

.	root of the DNS hierarchy
com	the com. top level domain
networklessons	the networklessons domain within .com
vps	the VPS hostname within domain networklessons.com

Now you have an idea what DNS is about. Let’s look at an actual example of a host that wants to find the IP address of a hostname. The host will send a DNS request and will receive a DNS reply from the server:



Once the host has the IP address, then it can try to contact the device it wants to communicate with. Below is the output of ipconfig on a Windows host:

```
C:\Users\Vmware>ipconfig /all | more

Windows IP Configuration

Host Name . . . . . : vmware
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : networklessons.local

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : networklessons.local
Description . . . . . : Intel(R) Ethernet Connection
I217-V
Physical Address. . . . . : 74-D0-2B-7D-22-8C
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

```
Link-local IPv6 Address . . . . . : 
fe80::dc21:558c:1574:96a2%3(Preferred)

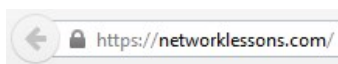
IPv4 Address. . . . . : 10.56.100.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.56.100.254
DHCPv6 IAID . . . . . : 259313707
DHCPv6 Client DUID. . . . . : 00-01-00-01-19-68-9A-AD-74-D0-2B-
7D-22-8C

DNS Servers . . . . . : 10.56.100.253
NetBIOS over Tcpi. . . . . : Enabled
```

You can see this host is using 10.56.100.253 as its DNS server. This is an internal DNS server on my local network.

! A DNS server on a local network typically only knows about hostnames and domain names on the local network. It can use a DNS server of the ISP or the root name servers when it has to find the IP addresses of hostnames on the Internet.

Let's open a webpage:



The host will now send a DNS request to find the IP address of networklessons.com, here's what it looks like in Wireshark:

```
Frame 4224: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface 0
Ethernet II, Src: AsustekC_7d:22:8c (74:d0:2b:7d:22:8c), Dst: Synology_3e:59:8b (00:11:32:3e:59:8b)
Internet Protocol Version 4, Src: 10.56.100.1 (10.56.100.1), Dst: 10.56.100.253 (10.56.100.253)
User Datagram Protocol, Src Port: 51012 (51012), Dst Port: 53 (53)
Domain Name System (query)
  [Response In: 4225]
  Transaction ID: 0x475b
  Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  Queries
    networklessons.com: type A, class IN
      Name: networklessons.com
      [Name Length: 18]
      [Label Count: 2]
      Type: A (Host Address) (1)
      Class: IN (0x0001)
```

Above you can see that the DNS query is sent to 10.56.100.253 using UDP port 53. The host wants to know the IP address for networklessons.com. Here's what the DNS server returns:

```
Frame 4225: 305 bytes on wire (2440 bits), 305 bytes captured (2440 bits) on interface 0
Ethernet II, Src: Synology_3e:59:8b (00:11:32:3e:59:8b), Dst: AsustekC_7d:22:8c (74:d0:2b:7d:22:8c)
Internet Protocol Version 4, Src: 10.56.100.253 (10.56.100.253), Dst: 10.56.100.1 (10.56.100.1)
User Datagram Protocol, Src Port: 53 (53), Dst Port: 51012 (51012)
Domain Name System (response)
  [Request In: 4224]
  [Time: 0.002551000 seconds]
  Transaction ID: 0x475b
  Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 13
  Additional RRs: 0
  Queries
    networklessons.com: type A, class IN
      Name: networklessons.com
      [Name Length: 18]
      [Label Count: 2]
      Type: A (Host Address) (1)
      Class: IN (0x0001)
  Answers
    networklessons.com: type A, class IN, addr 95.85.36.216
      Name: networklessons.com
      Type: A (Host Address) (1)
      Class: IN (0x0001)
      Time to live: 1300
      Data length: 4
      Address: 95.85.36.216 (95.85.36.216)
  Authoritative nameservers
```

Above you can see the answer, IP address 95.85.36.216 belongs to networklessons.com. The host can now use this IP address and try to establish a connection. If you want to take a look at these captures yourself then you can [Wireshark capture DNS Lookup](#).

I hope this has helped to give you an idea of how DNS works. If you have any questions, feel free to leave a comment.

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Command Line Interface (CLI)

🏷️ Tags: [Network Services](#)

Forum Replies



mauroaldair

Hello Rene,

Awesome even though i knew how DNS works, seeing on wireshark is just to awesome.
Well Done.

Mauro.



renjithpatterilmathe

Hi rene

the way of your teaching and explanation is awesome , and easy to understand everyone...
keep posted topics like this , looking for some security related stuffs as well...

Regards
Renjith P Mathew



ReneMolenaar

Hi Renjith,

Good to hear you like it! What security topics are you looking for?

Rene



renjithpatterilmathe

Hi rene
Looking for CCNA security level...





ReneMolenaar

Hi Renjith,

I have many tutorials that describe the CCNA security topics but right now I don't have a specific page with everything in order. Once I'm done with all the MPLS, multicast and QoS tutorials then I'll make a complete course for it.

Rene

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